

Week 2 Deliverable Documentation: Market & Competitor Study

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1. Competitor Analysis

As part of Week 2 research for Project StockSense, an in-depth market comparison was conducted. StockSense is evaluated against existing commercial smart-shelf solutions, standard weight-only smart bins, and traditional barcode or manual counting methods currently prevalent in Sri Lankan enterprises.

Feature / Metric	High-End Smart Shelves (e.g., Apex Actylus, Modula)	Standard Weight-Only Smart Bins	Traditional Barcode / Manual Counting	StockSense Solution (Proposed System)
Primary Sensing Tech	Weight + RFID + Computer Vision	Load Cell / Weight strain gauge only	Manual handheld barcode / QR scanner	Load Cell (HX711 24-bit ADC) + RFID (SPI/I ² C)
Environmental Monitoring	High-end custom options (expensive)	Rarely included	None	Integrated Temperature & Humidity (I ² C) per node
Hardware Price / Node	Very High (\$800 – \$2,500+ / LKR 240,000+)	Moderate (\$150 – \$300 / LKR 45,000+)	Low initial tool cost (Barcodes/Sheets)	Low-Cost (Target LKR 8,000–12,000 BOM / LKR 28,000 sale)
Offline Resilience	Cloud-reliant; requires active network	Basic offline buffer	N/A (Paper / Handheld)	Offline-first architecture (Local EEPROM/SD queue + auto sync)
Power Fault Tolerance	Requires uninterrupted grid / large UPS	Grid powered	Manual (No power needed)	Onboard Li-ion Battery Backup + Power Cut Outage Reporting
Language Support	English / Major global languages	English only	Manual input	Trilingual Dashboard (English, Sinhala, Tamil)
Analytics & AI	Standard ERP	Basic threshold alerts	Delayed manual	Local & Cloud AI

	integration / reorder rules		audits	(consumption forecasting, anomaly/theft alerts, run-out date prediction)
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Key Strategic Market Gaps Identified

1. Unadapted to Local Realities: Imported solutions cost upward of LKR 240,000 per node and fail during local power/internet drops. StockSense offers battery backup and offline queueing.
2. Environmental Vulnerability: Heat & humidity cause major stock degradation in local setups (pharmacies, garments, food). StockSense embeds climate monitoring into every node.
3. Usability & Pricing: High cost prevents SME adoption. StockSense targets an accessible price point (LKR 28,000) with a trilingual UI tailored for local shop-floor staff.

2. Customer Personas & Operational Use-Cases

2.1 Target Customer Personas

Persona: Dr. Nimanthi Jayasuriya – Chief Pharmacist & Clinic Owner

- **Pain Points:** Time-consuming manual counts, risk of expired vaccines/degraded heat-sensitive drugs, power cuts disrupting cloud tools.
- **Goals:** Real-time unit tracking, automated expiry/FEFO alerts, zero stock-outs of critical lifesaving medicines.
- **StockSense Alignment:** HX711 load cell unit counts, on-node temperature/humidity sensing, and 8+ hour battery backup.

Persona: Keshara Perera – Garment Factory Floor Supervisor

- **Pain Points:** Assembly downtime from sudden shortages of small parts (buttons, zippers, thread), skipped manual logging during rush shifts, line-side theft.
- **Goals:** Automated bin replenishment warnings before stock-outs, hands-free audit logs.
- **StockSense Alignment:** Continuous weight tracking triggers low-stock alerts; RFID tags identify restricted bins and batch material usage.

Persona: Dilshan Silva – Operations Manager, Retail Supermarket Chain

- **Pain Points:** Unannounced shelf stock-outs during peak hours, hours spent manually auditing shelves, floor staff struggling with English-only software.
- **Goals:** Live shelf visualization, predictive restock alerts, intuitive multilingual dashboard.
- **StockSense Alignment:** Smart-shelf weight telemetry updates local & cloud web dashboards continuously, supported by a trilingual UI (English, Sinhala, Tamil).

Persona: Ruwanthi Bandara – Cafe & Restaurant General Manager

- **Pain Points:** Unrecorded ingredient usage, internal shrinkage, food spoilage from storage temperature spikes, over-ordering perishables.
- **Goals:** Automate raw ingredient usage tracking and storage climate control.
- **StockSense Alignment:** Weight sensors calculate ingredient mass consumption; AI engine predicts run-out dates and alerts on abnormal usage spikes.

2.2 Top 10 Operational Use-Cases

1. Real-Time Automated Weight-to-Quantity Conversion

Primary Actor: Floor Worker / System Automated Trigger

1. Operator places batch items onto StockSense shelf bin.
2. HX711 ADC reads strain gauge differential voltage.
3. Firmware applies tare weight and average unit weight formula to derive exact item count.
4. Live count publishes over MQTT to dashboard.

2. Restock Threshold Alert & Reorder Prompt

Primary Actor: Pharmacy / Retail Manager

1. Stock falls below pre-configured safety threshold after items are removed.
2. StockSense detects step-drop in weight and evaluates threshold rule.
3. System dispatches push/SMS notification to manager with bin ID and required restock count.

3. Expiry Date & Batch Tracking via RFID Integration

Primary Actor: Storekeeper

1. New batch arrives with tagged RFID labels.
2. Storekeeper scans tag via MFRC522/PN532 reader prior to bin placement.
3. Firmware links tag UID with batch and expiry details in TimescaleDB.
4. System flags First-Expired, First-Out (FEFO) alerts if old batches remain.

4. Cold Chain / Storage Environment Excursion Alert

Primary Actor: Quality Assurance Officer

1. Integrated I²C temperature/humidity sensors monitor storage conditions.
2. Cooler door left open causes temperature to exceed >8°C for >10 mins.
3. Node triggers immediate high-priority alert on web dashboard and local buzzer.

5. Mains Power Outage Detection & Battery Failover

Primary Actor: Hardware Node / System Admin

1. Local grid power drops.
2. Power circuit switches seamlessly to internal Li-ion battery without MCU reset.
3. ESP32 detects grid loss, transmits 'Power Cut Event' MQTT payload, and enters power-save cycle (8+ hours uptime).

6. Store Network Disconnection & Offline Telemetry Queueing

Primary Actor: Firmware Network Manager

1. Wi-Fi router loses internet connection.
2. ESP32 switches state to Offline Mode and buffers timestamped weight/sensor logs to EEPROM/SD.
3. When connectivity restores, node flushes buffered payloads to Mosquitto broker sequentially without data loss.

7. Anomaly & Unaccounted Shrinkage/Theft Detection

Primary Actor: Store Manager / AI Engine

1. Unscheduled weight reduction occurs outside operating hours (e.g., 2:00 AM) or without POS transaction.
2. Python AI forecasting engine flags deviation as suspicious event.
3. Dashboard alerts security personnel with exact shelf ID and estimated loss quantity.

8. Multilingual Floor-Staff Inventory Lookup

Primary Actor: Local Floor Staff Member

1. Floor worker accesses web UI on a tablet in Sinhala or Tamil.
2. Worker selects SKU to view live bin location, stock quantity, and shelf map.
3. Performs fast audit without requiring technical training or English fluency.

9. AI Consumption Forecasting & Automated Reordering

Primary Actor: Supply Chain Manager

1. Machine learning models analyze historical telemetry in TimescaleDB.
2. Engine projects daily usage rates and run-out dates based on lead times.
3. System generates draft Purchase Orders (POs) automatically for supplier approval.

10. Multi-Node Calibration & Zero-Point Tare Setup

Primary Actor: Field Hardware Technician

1. Technician attaches new load cell platform to node.
2. Triggers calibration routine via web UI or physical setup button.
3. Node saves zero-offset (empty bin), prompts for known test weight (e.g., 1000g), calculates calibration factor, and saves to flash storage.

3. Customer Survey Preparation

To validate field requirements and test pricing/feature assumptions directly against real-world users, a short 5-question interview script was prepared for conducting calls/interviews with 5 local target business owners.

Survey Protocol Overview

Target Interviewees: Store Managers, Operations Supervisors, Pharmacy Owners, Factory Supervisors

Estimated Time: 5–7 Minutes per session

Primary Objective: Gather feedback on stock-out costs, shrinkage, power outage frequencies, and target price feasibility.

3.1 Call & Survey Script Template

Opening Script / Introduction:

"Hello Kasun, my name is Kavindu from Codexus Technologies. We are developing StockSense, an automated smart inventory and shelf monitoring solution designed specifically for local businesses. I am reaching out to understand how your team currently manages stock and where the biggest operational headaches occur. This will take less than 5 minutes. Do you have a moment?"

3.2 5 Core Survey Questions

Question 1: Manual Counting & Audit Frequency

"How do you currently track inventory levels, and how much time does your staff spend performing manual stock counts every week?"

- **Primary Objective:** Quantify labor costs and time wasted on manual audits vs. automated weight/RFID monitoring.
- **Follow-up Probe:** Do you experience human errors or unrecorded stock usage during peak working hours?

Question 2: Business Impact of Stock-outs & Discrepancies

"How often do unexpected stock-outs or stock discrepancies occur, and what is the direct financial or operational cost to your business?"

- **Primary Objective:** Validate the urgency of real-time low-stock alerts and weight-to-quantity tracking.
- **Follow-up Probe:** Have you ever lost customers or delayed line production due to missing items?

Question 3: Infrastructure Constraints (Power Cuts & Connectivity)

"When grid power cuts or Wi-Fi disconnections occur at your facility, how are your operations affected, and what backup systems do you rely on?"

- **Primary Objective:** Validate the necessity of StockSense's 8-hour battery backup and offline-first telemetry queueing.
- **Follow-up Probe:** Would offline queueing and power-failure reporting be critical features for your setup?

Question 4: Cold-Chain & Environmental Storage Sensitivity

"Do you handle items sensitive to temperature or humidity (e.g., medicines, raw food items, textiles), and how do you monitor storage conditions?"

- **Primary Objective:** Justify integrating onboard temperature/humidity sensors directly into the smart bin hardware.
- **Follow-up Probe:** Have you ever experienced stock spoilage due to undetected climate fluctuations or open storage doors?

Question 5: Adoption Feasibility & Target Price Sensitivity

"If a smart system automatically tracked item counts, predicted restock dates, and sent instant alerts, what price range would make sense for your business per smart shelf/bin unit?"

- **Primary Objective:** Validate market viability against StockSense's target price point (LKR 28,000 / ~\$85 USD vs. expensive \$800+ global imports).
- **Follow-up Probe:** Would your floor staff benefit from a dashboard available in local languages (Sinhala/Tamil)?

3.3 Feedback Log Sheet Template

The following log table will be used to record findings from the 5 target business interviews:

Business Name & Sector	Contact Person	Current Counting Method	Main Pain Point	Max Target Price / Node	Notes / Feature Requests
City Pharmacy	Dr. Perera	Nightly manual log	Cold chain & unrecorded counts	LKR 25,000–30,000	Wants Sinhala UI & SMS alerts
Apex Apparel Ltd	K. Wickramasinghe	Handheld barcode / Paper	Line downtime due to missing buttons	LKR 30,000–35,000	Needs fast RFID batch scan
Central Cafe	M. Fernando	Weekly manual audit	Perishable ingredient shrinkage	LKR 20,000–25,000	Temperature graph on dashboard
Target 4					
Target 5					